

Inversa de un morfismo de anillos biyectivo

Ejercicio 1. Sea $f: R \rightarrow S$ un morfismo de anillos biyectivo, demuestra que $f^{-1}: S \rightarrow R$ es un morfismo de anillos.

Solución: Sean $a, b \in S$, como f es sobreyectiva, $\exists x, y \in R : f(x) = a \wedge f(y) = b$. Comprobemos las tres propiedades de la **morfismo-anillos**/Definición 1.

$$(i) \quad f^{-1}(a + b) = f^{-1}(f(x) + f(y)) = f^{-1}(f(x + y)) = x + y = f^{-1}(a) + f^{-1}(b).$$

$$(ii) \quad f^{-1}(a \cdot b) = f^{-1}(f(x) \cdot f(y)) = f^{-1}(f(x \cdot y)) = x \cdot y = f^{-1}(a) \cdot f^{-1}(b).$$

$$(iii) \quad \text{Como } f \text{ es un morfismo de anillos, } f(1_R) = 1_S. \text{ Luego, } f^{-1}(1_S) = f^{-1}(f(1_R)) = 1_R.$$

Por tanto, f^{-1} es un morfismo de anillos. ■

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